

Graduate-Preparation Roadmap: PDEs, Compressible Flow, and a Supersonic Inlet Solver From Conservation Laws to Real-Gas Hypersonics

Fall 2026 – Preparation for Graduate Study in Aerospace Engineering

1 Purpose and Scope of This Revision

The original nine-problem roadmap (advection $\rightarrow$ Burgers $\rightarrow$ potential flow $\rightarrow$ 1D Euler $\rightarrow$ shock tube $\rightarrow$ quasi-1D nozzle $\rightarrow$ 2D Euler $\rightarrow$ wedge $\rightarrow$ inlet) is the correct *skeleton*, but it stops at first-order, inviscid, calorically-perfect-gas CFD. That is undergraduate-terminal material. It does not touch four things a graduate aerodynamics/propulsion program will assume you can already reason about on day one:

1. **Rigor of the hyperbolic theory itself** – why weak solutions exist and are (or are not) unique, what the Lax and Oleinik entropy conditions actually say, and why the Lax–Wendroff theorem is the reason finite-volume methods are trusted at all.
2. **Numerical method quality** – first-order upwind and Rusanov fluxes are diffusive enough to smear the exact shock structure you are trying to study. Graduate-level work expects Roe/HLLC approximate Riemann solvers, MUSCL reconstruction, TVD limiters, and an honest discussion of why Godunov’s theorem forbids a linear, monotone scheme above first order.
3. **Viscosity and turbulence**, not as an afterthought but as the physics that actually determines inlet performance: boundary-layer separation, shock/boundary-layer interaction, and the closure problem of Reynolds-averaged turbulence.
4. **Real-gas and hypersonic effects** – at the Mach numbers your target courses operate in (hypersonic aerodynamics, high-temperature gas dynamics, advanced airbreathing propulsion), the calorically-perfect ideal gas model breaks down. You should understand *why* and *when*.

This revision keeps the original inviscid backbone (Levels 1–7 below map onto the original Problems 1–7) but raises the required rigor at every level, and adds five new levels (8–12) that were previously waved at in a single closing section. It is built specifically around six graduate courses at UMD that you have flagged as of interest, so that the roadmap doubles as a self-study on-ramp into each one.

1.1 Course Mapping

Course	Title	Roadmap levels that prepare you for it
ENAE674	Aerodynamics of Compressible Fluids	Levels 4–7 (1D/2D Euler, characteristics, shock/expansion theory)
ENAE676	Turbulence	Level 9 (Reynolds averaging, closure, scales of motion)
ENAE682	Hypersonic Aerodynamics	Levels 11–12 (Newtonian theory, hypersonic similarity, viscous interaction)
ENAE683	High Temperature Gas Dynamics	Level 11 (thermally/chemically reacting flow, equilibrium and nonequilibrium gas)
ENAE685	Computational Fluid Dynamics II	Levels 8, 10 (high-order finite-volume schemes, viscous discretization)
ENAE665	Advanced Airbreathing Propulsion	Level 13 (inlet performance, engine/airframe integration)

Working through Levels 1–13 in order is, in effect, a self-taught prerequisite pass for all six courses. None of this replaces the courses – it means you walk into them already fluent in the governing equations instead of meeting them for the first time in lecture.

2 Mathematical Prerequisites (Do Not Skip)

Before Level 1, you should be able to state and use the following without looking them up. If any are unfamiliar, spend a week on them first; everything downstream depends on them.

1. **PDE classification.** For a second-order linear PDE $Au_{xx} + 2Bu_{xy} + Cu_{yy} + \dots = 0$, classify by the sign of $B^2 - AC$ and connect this to the number and nature of real characteristic directions. Know why the steady 2D Euler equations are elliptic for subsonic flow and hyperbolic for supersonic flow, and why this changes the required boundary conditions and the domain of dependence.
2. **Method of characteristics**, general case: for a first-order quasilinear PDE $a(x, t, u)u_x + b(x, t, u)u_t = c(x, t, u)$, derive the characteristic ODEs $\frac{dx}{ds} = a$, $\frac{dt}{ds} = b$, $\frac{du}{ds} = c$, and be able to construct solutions by characteristic tracing for both linear and quasilinear cases.
3. **Weak solutions.** Derive the weak form of a conservation law $\int \int (u \phi_t + f(u) \phi_x) dx dt = 0$ for all test functions $\phi \in C_0^\infty$, and explain precisely why weak solutions are required once characteristics cross.
4. **Well-posedness in the Hadamard sense:** existence, uniqueness, continuous dependence on data. Know one example of an ill-posed problem (e.g., backward heat equation) so “well-posed” is not just a word.
5. **Linear algebra of hyperbolic systems:** for $\mathbf{U}_t + \mathbf{A}\mathbf{U}_x = 0$, hyperbolicity means $\mathbf{A}$ is real-diagonalizable. Be fluent with eigenvalues, right eigenvectors, characteristic variables $\mathbf{U} \rightarrow$ diagonalized system, and Riemann invariants along characteristics.
6. **Basic real analysis of numerical schemes:** consistency, stability, and convergence, and the statement of the **Lax Equivalence Theorem** (consistency + stability $\Leftrightarrow$ convergence, for well-posed linear problems). Know what von Neumann stability analysis actually bounds.

7. **Godunov’s theorem:** no linear, monotonicity-preserving scheme can be higher than first-order accurate. This is the reason nonlinear (limiter-based) high-resolution schemes exist at all – understand the proof sketch, not just the statement.

3 Level-by-Level Roadmap

Lvl	PDE / System	Graduate-level lesson	Rigor
1	Linear advection	Characteristics, Lax equivalence, modified-equation analysis	Undergrad+
2	Inviscid Burgers	Weak solutions, Lax/Oleinik entropy conditions, viscosity limit	Grad-entry
3	Laplace / potential flow	Elliptic well-posedness, Green’s functions, transonic small-disturbance limit	Grad-entry
4	1D Euler	Hyperbolicity proof, characteristic variables, Riemann invariants	Grad-core
5	Exact + approximate Riemann solvers	Roe linearization, HLLC contact restoration	Grad-core
6	Quasi-1D Euler w/ friction & heat	Fanno/Rayleigh flow, choking, unsteady nozzle startup	Grad-core
7	2D Euler, oblique shocks, P–M fans	Method of characteristics for supersonic design	Grad-core
8	High-order finite volume	MUSCL, TVD limiters, WENO, Godunov’s theorem in practice	Grad-advanced
9	RANS turbulence closure	Reynolds averaging, closure problem, k – ε /SST	Grad-advanced
10	Compressible Navier–Stokes	Compressible boundary layers, Crocco–Busemann, SBLI	Grad-advanced
11	Real-gas thermochemistry	Thermally/chemically reacting flow, equilibrium vs. nonequilibrium	Grad-advanced
12	Hypersonic flow theory	Newtonian theory, hypersonic similarity, viscous interaction parameter	Grad-advanced
13	Inlet performance & integration	Total-pressure recovery, mass capture, engine/airframe coupling	Capstone

3.1 Level 1: Linear Advection – Numerical Analysis, Not Just Implementation

Govern by $u_t + au_x = 0$, $u(x, 0) = u_0(x)$, with solution $u(x, t) = u_0(x - at)$ by characteristics.

Required work.

1. Derive the exact solution by the method of characteristics and state its domain of dependence and range of influence.
2. Implement first-order upwind, Lax–Wendroff, and Beam–Warming schemes for the same initial data.
3. Perform a von Neumann stability analysis for each scheme: substitute $u_j^n = \hat{u}^n e^{ikj\Delta x}$ and derive the amplification factor $g(k)$. Show algebraically why upwind requires $C \leq 1$ and why Lax–Wendroff’s stability bound differs.
4. Derive the **modified equation** for upwind (i.e., the PDE the discrete scheme actually solves to leading order, including a diffusive term $\propto a\Delta x(1 - C)/2 u_{xx}$). Use it to explain numerical diffusion quantitatively, not just descriptively.
5. Derive the modified equation for Lax–Wendroff and show its leading error term is dispersive

(u_{xxx}) rather than diffusive – explain why this produces oscillations near discontinuities instead of smearing.

6. State the Lax Equivalence Theorem and verify empirically (via a refinement study) that your scheme’s observed convergence rate matches its truncation-error order.

Checkpoint question. Why does a scheme that is stable and consistent for smooth data still fail to converge to the entropy-satisfying solution once you introduce a discontinuity? (This motivates Level 2.)

3.2 Level 2: Inviscid Burgers – Entropy, Uniqueness, and the Vanishing-Viscosity Limit

Govern by $u_t + u u_x = 0 \Leftrightarrow u_t + \left(\frac{u^2}{2}\right)_x = 0$.

Required work.

1. Derive the Rankine–Hugoniot condition $s = \frac{f(u_R) - f(u_L)}{u_R - u_L}$ from the *integral* form of the conservation law across a moving control volume straddling the discontinuity – do not just quote the formula.
2. Solve the shock problem ($u_L = 2, u_R = 1$) and the rarefaction problem ($u_L = 1, u_R = 2$); construct the self-similar rarefaction fan $u(x, t) = x/t$ for $u_L < x/t < u_R$.
3. State the **Lax entropy condition** $f'(u_L) > s > f'(u_R)$ and the more general **Oleinik entropy condition**, and show that the naive “other” weak solution to the rarefaction problem (an unphysical expansion shock) satisfies Rankine–Hugoniot but violates entropy.
4. Derive the **viscous Burgers equation** $u_t + uu_x = \nu u_{xx}$, construct its traveling-wave (Taylor shock) solution, and show that as $\nu \rightarrow 0$ this profile converges to the entropy-satisfying weak solution – this is the vanishing-viscosity selection principle, and it is the theoretical justification for why artificial/numerical viscosity in a CFD scheme is not merely a numerical trick but a physically motivated regularization.
5. Implement a first-order Godunov scheme using the exact Riemann solver for Burgers’ equation, and verify shock speed and rarefaction fan width converge to the analytic solution under grid refinement.

3.3 Level 3: Laplace Equation, Potential Flow, and the Transonic Limit

Required work.

1. Derive $\nabla^2 \phi = 0$ from incompressibility ($\nabla \cdot \mathbf{u} = 0$) and irrotationality ($\nabla \times \mathbf{u} = 0$), and prove well-posedness of the Dirichlet problem via an energy (uniqueness) argument: assume two solutions, subtract, apply Green’s first identity to the difference, and show the Dirichlet energy vanishes.
2. Solve a rectangular boundary-value problem analytically by separation of variables and Fourier series; verify against a finite-difference solver.
3. Derive the **Prandtl–Glauert equation** $(1 - M_\infty^2)\phi_{xx} + \phi_{yy} = 0$ from the full-potential equation by linearizing about a uniform freestream, and explain why this equation changes type (elliptic $\rightarrow$ hyperbolic) exactly at $M_\infty = 1$ – connect this directly to your Section 2 classification work.
4. Explain concretely (in terms of vorticity generation across curved shocks, and in terms of the entropy jump across a shock) why potential-flow theory cannot represent a shock of finite

strength, and why it is nonetheless still used as a design tool for weak-shock transonic and supersonic problems.

3.4 Level 4: 1D Compressible Euler Equations – Full Derivation and Hyperbolicity Proof

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = 0, \quad \mathbf{U} = \begin{pmatrix} \rho \\ \rho u \\ E \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(E + p) \end{pmatrix}, \quad E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho u^2.$$

Required work.

1. Derive mass, momentum, and energy conservation each from a differential control volume (Reynolds transport theorem), and separately from the integral form on a fixed control volume, showing the two are equivalent under sufficient smoothness.
2. Close the system with the ideal-gas equation of state and derive E in terms of conservative variables.
3. Compute the flux Jacobian $A = \partial \mathbf{F} / \partial \mathbf{U}$ explicitly (all nine entries in terms of u , H (specific total enthalpy), and γ), and **prove** A is real-diagonalizable by computing its eigenvalues $\lambda = u - a$, u , $u + a$ and a complete set of right eigenvectors – this is the actual hyperbolicity proof, not an assertion.
4. Derive the **characteristic variables** (Riemann invariants for the linearized/isentropic case) $J_{\pm} = u \pm \frac{2a}{\gamma - 1}$ and show they are constant along the $C_{\pm}$ characteristics for smooth isentropic flow.
5. Give the physical interpretation of each eigenvalue (entropy/contact wave advecting at u ; left- and right-running acoustic waves at $u \mp a$) and connect this to subsonic vs. supersonic boundary-condition counting (how many physical boundary conditions are needed at an inflow/outflow boundary, and why it depends on the sign pattern of the eigenvalues there).

3.5 Level 5: The Riemann Problem – Exact and Approximate Solvers

Required work.

1. Solve the Sod shock tube *exactly*: set up the star-region equations (pressure and velocity across the contact discontinuity), solve the nonlinear pressure equation iteratively (Newton–Raphson), and reconstruct ρ , u , p in all five regions (left state, left rarefaction fan, star-left, star-right, right state, right state – rarefaction → contact → shock).
2. Derive the **Roe linearization**: construct the Roe-averaged state $\tilde{u}, \tilde{H}, \tilde{a}$ using the density-weighted (Roe) averaging that makes the linearized flux Jacobian $\tilde{A}$ exactly satisfy $\mathbf{F}_R - \mathbf{F}_L = \tilde{A}(\mathbf{U}_R - \mathbf{U}_L)$, and explain why this property matters for exact single-shock capturing.
3. Implement the Roe flux and identify the **entropy fix** (e.g., Harten’s) needed to prevent unphysical expansion shocks at sonic points – reproduce the failure mode without the fix, then fix it.
4. Derive the **HLLC** flux: start from the two-wave HLL flux, then show how restoring an intermediate contact/shear wave (with pressure and normal velocity continuous across it) recovers the contact discontinuity that HLL smears.

5. Compare Rusanov, Roe, and HLLC on the Sod problem and a strong shock/blast-wave problem; quantify contact-discontinuity smearing (number of cells to resolve the contact within a tolerance) as a function of scheme.

3.6 Level 6: Quasi-1D Nozzle – Choking, Friction, and Heat Addition

$$\frac{\partial(A\rho)}{\partial t} + \frac{\partial(A\rho u)}{\partial x} = 0, \quad \frac{\partial(A\rho u)}{\partial t} + \frac{\partial[A(\rho u^2 + p)]}{\partial x} = p \frac{dA}{dx}, \quad \frac{\partial(AE)}{\partial t} + \frac{\partial[Au(E + p)]}{\partial x} = 0.$$

Required work.

1. Derive the area–Mach relation $\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}$ from the quasi-1D governing equations directly (not by lookup), and derive the area-velocity relation $\frac{dA}{A} = (M^2 - 1) \frac{du}{u}$ to explain why subsonic and supersonic nozzles behave oppositely at a converging/diverging section.
2. Derive **Fanno flow** (adiabatic flow with friction, constant area) and **Rayleigh flow** (frictionless flow with heat addition, constant area) as two independent limiting cases, and derive the choking condition ($Mach \rightarrow 1$) for each.
3. Numerically sweep back pressure to obtain: fully subsonic flow, choked flow, supersonic acceleration past the throat, an internal normal shock at a numerically located position, and a fully supersonic exit; verify the shock location against the exact quasi-1D shock relations.
4. Simulate unsteady nozzle startup (impulsive pressure ratio applied at $t = 0$) and identify the transient shock/expansion wave system before it settles to steady state – this is your first genuinely unsteady compressible-flow simulation.

3.7 Level 7: 2D Euler, Oblique Shocks, and the Method of Characteristics

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x + \mathbf{G}(\mathbf{U})_y = 0, \quad \mathbf{U} = (\rho, \rho u, \rho v, E)^T.$$

Required work.

1. Derive the oblique-shock relations from the 1D normal-shock relations applied to the shock-normal velocity component only, and derive the θ – β – M relation $\tan \theta = 2 \cot \beta \frac{M_\infty^2 \sin^2 \beta - 1}{M_\infty^2 (\gamma + \cos 2\beta) + 2}$ from first principles.
2. Derive the **Prandtl–Meyer function** $\nu(M)$ from the isentropic simple-wave relations for a centered expansion, and use it to compute deflection angles for a supersonic expansion corner.
3. Implement the classical **method of characteristics (MOC)** for steady 2D supersonic flow: construct the C_+/C_- characteristic mesh from the compatibility equations, and design a minimum-length nozzle contour using MOC – this is the standard graduate design tool this level should leave you fluent in, and it is the analytic backbone against which you verify your 2D Euler solver.
4. Verify your 2D Euler solver on the wedge problem ($M_\infty = 2.0$, $\theta = 10^\circ$, $\gamma = 1.4$): report β_{CFD} , β_{theory} , percent error, and post-shock pressure ratio and Mach number against oblique-shock theory. Do not proceed to Level 13 until this error is small and grid-converged.

3.8 Level 8: High-Order Finite-Volume Methods

Required work.

1. State and prove (or carefully reproduce the proof of) **Godunov's theorem**: no linear scheme that is monotonicity-preserving can be more than first-order accurate. Explain precisely why this forces high-resolution schemes to be nonlinear (solution-dependent) even for a linear PDE.
2. Implement **MUSCL reconstruction** with at least two slope limiters (minmod and van Leer), and show the resulting scheme is formally second-order in smooth regions while remaining non-oscillatory at discontinuities.
3. Derive the **TVD (total variation diminishing)** condition on a limiter function and show minmod and van Leer satisfy it; plot the limiter functions against the TVD admissible region on a Sweby diagram.
4. Implement at least a 3rd-order **WENO** reconstruction (WENO5) and compare against MUSCL on a problem with both a smooth extremum and a shock (e.g., Shu–Osher problem) – this is the standard test case for demonstrating a scheme resolves both smooth structure and discontinuities without excessive clipping or oscillation.
5. Perform a formal order-of-accuracy study (log-log error vs. Δx) for each scheme on a smooth manufactured solution, and confirm the observed slopes match the design order.

3.9 Level 9: Turbulence – Reynolds Averaging and Closure

Required work.

1. Derive the **Reynolds-averaged Navier–Stokes (RANS)** equations by Reynolds decomposition $u_i = \bar{u}_i + u'_i$ and time-averaging the Navier–Stokes equations; identify the **Reynolds stress tensor** $-\rho \overline{u'_i u'_j}$ as the new unknown that makes the system unclosed – state explicitly why this is called the closure problem.
2. Derive the **Boussinesq eddy-viscosity hypothesis** and explain its physical assumptions and known failure modes (streamline curvature, strong adverse pressure gradients, separated flow).
3. Derive the exact (unclosed) turbulent kinetic energy equation from the RANS momentum equation, and show how it motivates the transport-equation structure of a k – ε or k – ω model.
4. Compare (conceptually and, if time allows, numerically on a flat-plate boundary layer) a zero-equation (mixing length), two-equation (k – ε), and k – ω SST closure – explain why SST blends near-wall k – ω with freestream k – ε behavior.
5. Discuss why RANS is currently the workhorse for inlet design despite its known deficiencies, and what LES/DES buys you at what cost, so you can justify a modeling choice in a design report rather than default to it.

3.10 Level 10: Compressible Navier–Stokes and Viscous Interactions

$$\mathbf{U}_t + \nabla \cdot \mathbf{F}_c = \nabla \cdot \mathbf{F}_v, \quad \tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \nabla \cdot \mathbf{u} \right), \quad \mathbf{q} = -k \nabla T.$$

Required work.

1. Derive the viscous stress tensor from Stokes' hypothesis and incorporate it, along with Fourier conduction, into the full compressible Navier–Stokes system; identify which terms vanish in the inviscid (Euler) limit and confirm your Level 4–7 equations are recovered.

2. Derive the **compressible boundary-layer equations** by an order-of-magnitude (boundary-layer) scaling argument applied to the full Navier–Stokes equations, and derive the **Crocco–Busemann relation** linking the temperature and velocity profiles in a compressible boundary layer with recovery temperature/Prandtl number effects.
3. Explain the mechanism of **shock/boundary-layer interaction (SBLI)**: adverse pressure gradient imposed by the shock foot, boundary-layer thickening, and (for strong enough interaction) separation with a lambda-shock foot – this is directly relevant to inlet unstart.
4. Numerically discretize the viscous flux terms (central differencing is standard, unlike the convective terms) and validate your viscous solver against the Blasius flat-plate boundary-layer solution (compare $\delta(x)$, skin friction $C_f(x)$ against the incompressible Blasius result, then against the compressible correction).

3.11 Level 11: Real-Gas and High-Temperature Effects

Required work.

1. Explain, with the relevant statistical-thermodynamic argument (equipartition and vibrational mode excitation), why $\gamma = 1.4$ fails as post-shock temperature rises, and derive the **thermally perfect gas** model ($p = \rho RT$, but $c_v(T)$, $c_p(T)$ temperature-dependent) as the first correction.
2. Distinguish **chemical equilibrium**, **chemical nonequilibrium**, and **thermal (vibrational) nonequilibrium** flow, and state the governing Damköhler-number argument for which regime applies at a given flight condition.
3. Derive the normal-shock jump relations for a thermally perfect (variable- γ) gas and compare post-shock temperature and pressure against the calorically perfect result at a hypersonic Mach number (e.g., $M_\infty = 8$) – quantify the error from assuming constant γ .
4. Describe (conceptually, with the governing rate equations) how a finite-rate chemistry model (e.g., a 5- or 11-species air model) modifies the Euler/Navier–Stokes system with additional species continuity equations and source terms, without necessarily implementing a full reacting solver.

3.12 Level 12: Hypersonic Aerodynamics

Required work.

1. Derive **Newtonian impact theory** ($C_p = 2 \sin^2 \theta$) from momentum flux arguments and state its regime of validity; derive the **modified Newtonian** correction using the stagnation pressure coefficient.
2. Derive the **hypersonic small-disturbance similarity parameter** $K = M_\infty \tau$ (where τ is a characteristic thickness ratio) and explain why hypersonic flows over different bodies collapse onto common pressure curves when plotted against K .
3. Derive the **hypersonic viscous interaction parameter** $\bar{\chi} = M_\infty^3 \sqrt{C}/\sqrt{Re_x}$ and explain the strong- and weak-interaction regimes it delineates for a hypersonic boundary layer.
4. Compute the hypersonic shock-layer thickness and post-shock conditions for a blunt body using both Newtonian theory and exact oblique-shock theory in the hypersonic limit, and compare.
5. Connect Levels 11 and 12: explain, for a given flight Mach number and altitude, how you would decide whether a calorically perfect, thermally perfect, or reacting-flow model is required for an inlet analysis to be credible.

3.13 Level 13 (Capstone): Supersonic/Hypersonic Inlet Performance and Integration

Only attempt this after the wedge (Level 7) and, ideally, a viscous verification case (Level 10) are grid-converged and validated.

Required work.

1. Build the inlet geometry (external-compression ramp, or mixed-compression as a stretch goal) and sweep $M_\infty = 1.5, 2.0, 2.5, 3.0, 4.0$ with varying ramp angle and contraction ratio.
2. Report p/p_∞ , ρ/ρ_∞ , $M(x, y)$, and stagnation-pressure ratio field $p_0/p_{0,\infty}$; compute the **total-pressure recovery** $\pi_d = p_{0,\text{exit}}/p_{0,\infty}$ and compare against the MIL-E-5008B empirical standard for the corresponding freestream Mach number.
3. Quantify **mass capture** and spillage as a function of ramp angle and back pressure, and identify the onset of **inlet unstart** (loss of supersonic flow swallowing, typically triggered by a shock/boundary-layer separation event – connect directly to your Level 10 SBLI work).
4. Investigate shock train / shock-train interaction if a mixed-compression or isolator section is included, and discuss (qualitatively, referencing Level 9) where RANS turbulence-model choice most strongly affects your separation prediction.
5. Frame the final engineering question explicitly: how does inlet geometry trade compression efficiency against total-pressure loss and off-design robustness, and what would you change about the design based on your swept results?

4 Repository Structure

```
supersonic-inlet-cfd/  
|-- src/  
|   |-- advection_1d.py  
|   |-- burgers_1d.py  
|   |-- potential_flow_2d.py  
|   |-- euler_1d.py  
|   |-- riemann_exact.py  
|   |-- riemann_roe.py  
|   |-- riemann_hllc.py  
|   |-- nozzle_q1d.py  
|   |-- muscl_weno.py  
|   |-- euler_2d.py  
|   |-- supersonic_wedge.py  
|   |-- moc_nozzle.py  
|   |-- navier_stokes_2d.py  
|   |-- turbulence_ke_komega.py  
|   |-- real_gas_thermo.py  
|   '-- inlet_solver.py  
|  
|-- docs/  
|   |-- governing_equations.pdf  
|   |-- numerical_methods.pdf  
|   |-- turbulence_closure.pdf
```

```
| |-- real_gas_notes.pdf
| |-- verification.pdf
|
|-- results/
| |-- shock_tube/
| |-- nozzle/
| |-- wedge/
| |-- boundary_layer/
| |-- inlet/
|
|-- data/
|-- geometry/
'-- README.md
```

5 Comprehensive Problem Set (Qualifying-Exam Style)

Work these after finishing the corresponding level, closed-book, to check whether the derivation actually transferred rather than just being followed along with.

1. Prove that the 1D Euler flux Jacobian is diagonalizable and give the full eigenvector matrix in terms of u , a , H .
2. A weak solution of Burgers' equation satisfies Rankine–Hugoniot but violates the Lax entropy condition. Construct such a solution explicitly and explain, using the vanishing-viscosity argument, why it is unphysical.
3. Derive the Roe-averaged sound speed $\tilde{a}$ in terms of $\tilde{H}$ and $\tilde{u}$ and show the resulting linearized Jacobian satisfies the property $\mathbf{F}_R - \mathbf{F}_L = \tilde{A}(\mathbf{U}_R - \mathbf{U}_L)$ exactly for the Euler equations.
4. Starting from the quasi-1D momentum equation, derive $dA/A = (M^2 - 1) du/u$ and use it to explain, in one physically complete paragraph, why a converging duct accelerates subsonic flow but decelerates supersonic flow.
5. State Godunov's theorem precisely and explain why a limiter function must be a nonlinear function of the local solution gradient in order to be both high-order and non-oscillatory.
6. Derive the Reynolds stress term that appears when you Reynolds-average the nonlinear convective term in the Navier–Stokes momentum equation, and explain in what sense this makes RANS an unclosed system.
7. For a normal shock at $M_\infty = 8$ in air, estimate the error introduced by assuming $\gamma = 1.4$ is constant through the shock, using a thermally perfect gas correction; state your assumptions.
8. Derive the hypersonic viscous interaction parameter and explain what physically distinguishes the weak- and strong-interaction regimes for a hypersonic flat-plate boundary layer.
9. Given a computed inlet flow field, define total-pressure recovery and mass capture ratio, and explain how each would appear in a shock/boundary-layer-driven unstart event.
10. Explain, end to end, why the sequence advection $\rightarrow$ Burgers $\rightarrow$ 1D Euler $\rightarrow$ 2D Euler $\rightarrow$ Navier–Stokes is not arbitrary pedagogy but tracks the actual mathematical structure (linear $\rightarrow$ nonlinear scalar $\rightarrow$ nonlinear hyperbolic system $\rightarrow$ multi-dimensional system $\rightarrow$ mixed hyperbolic-parabolic system) that a production CFD code must handle.

6 Minimum Standard Before Calling the Project Graduate-Ready

You should be able to do all of the following without hand-waving or looking anything up:

1. Derive the compressible Euler equations from conservation principles and prove hyperbolicity via the flux Jacobian eigen-decomposition.
2. Derive the Rankine–Hugoniot condition and state and apply the Lax/Oleinik entropy conditions.
3. Derive the Roe average and the HLLC flux, and explain when each fails (entropy-violating expansion shocks; contact smearing).
4. State Godunov’s theorem and explain why MUSCL/WENO reconstruction is a nonlinear workaround, not a loophole.
5. Derive the RANS equations and explain, using the Reynolds stress term, exactly why turbulence modeling is a closure problem rather than a discretization problem.
6. Derive the compressible boundary-layer equations and explain the mechanism of shock/boundary-layer interaction and inlet unstart in those terms.
7. Explain, quantitatively for at least one Mach number, why a calorically perfect gas model breaks down and what a thermally perfect or reacting-flow correction changes.
8. Predict an oblique-shock angle analytically and recover it numerically to within a small, grid-converged error.
9. Report total-pressure recovery and mass capture for a swept-Mach-number inlet study, and explain the compression/loss trade-off in your own words, not the roadmap’s.
10. Justify, for your inlet geometry and flight condition, every modeling choice you made (gas model, turbulence closure, numerical scheme) rather than defaulting to the first option covered.

7 Bottom Line

The original roadmap got you to a first-order inviscid Euler solver – a solid but undergraduate-terminal stopping point. Graduate aerodynamics and propulsion coursework assumes fluency with weak-solution theory and entropy conditions, high-resolution shock-capturing schemes, Reynolds-averaged turbulence closure, viscous shock/boundary-layer interaction, and real-gas thermochemistry at hypersonic conditions. Levels 8–13 above are not optional extensions; they are the material that separates a solver that reproduces a textbook oblique-shock angle from one whose inlet-performance predictions you could defend in front of the faculty teaching ENAE674, ENAE676, ENAE682, ENAE683, and ENAE665.